

ARPIT KUMAR

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EDUCATION

B.Tech CSE (AI & ML) | *IMS Engineering College, Ghaziabad*
CGPA: 7.04/10

2022–2026

Senior Secondary, CBSE | *Greenfields Public School, Delhi*
Stream: PCM Cs | Percentage: 84.4%

2022

EXPERIENCE

AI/LLM Solutions Developer | *Botter Solutions Pvt. Ltd. — Rovabot.io*

Aug 2025 – Nov 2025

- Researched OCR and Classification models; fine-tuned, trained, and tested LLMs, VLMs, and vLLMs including LayoutLMV3, QWEN3, CascadeTabNet, TableLLM, and PaddleOCR-VL.
- Managed dataset preparation and annotations using CVAT.
- Utilised cloud GPUs (RunPod.io) and handled cloud resource management.
- Consulted and assisted in problem-solving, workflow formulation, and project management.

Data Annotation Specialist & ML Assistant | *DraftTimeAI — Drafttime.com*

Feb 2026 – April 2026

- Annotated complex architectural diagrams on CVAT; interpreted ambiguous visual inputs and maintained labeling consistency.
- Assisted in research and model training using Detectron2 and YOLOv8x.
- Analyzed visual and contextual data to identify inconsistencies and improve annotation quality.
- Collaborated in refining dataset structure to enhance model training performance.

PROJECTS

Synapse: A Multimodal Assistive Reading System

Python, Typescript, TensorFlow, PyTorch, OpenCV

Paper Accepted: Scopus IEEE Xplore NMIC 2026

- Co-authored "Synapse," a privacy-first, browser-based assistive reading platform for dyslexic users.
- Built a serverless attention-tracking pipeline using MediaPipe for real-time, hardware-free head and iris tracking.
- Trained an XGBoost cognitive difficulty model on the ZuCo dataset, achieving 0.9989 AUC via behavioral gaze features.
- Integrated word-level scaffolding, sentence simplification, and text-to-speech within a unified React/TypeScript architecture.

VidVision3D — Markerless Motion Capture System

Python, Flask, BlazePose, React, Unity 3D

- Engineered a low-cost motion capture system using computer vision and real-time video processing.
- Extracted 3D landmarks (body, hands, pose) from video using MediaPipe and BlazePose.
- Integrated Flask backend with React frontend for live capture and export; reduced animation production costs by **80%** for independent creators.

Multimodal Emotion Recognition System

Python, TensorFlow, PyTorch, OpenCV, HuggingFace

- Engineered a real-time AI system to detect emotions by fusing voice, facial, and textual signals.
- Trained custom CNN on RAVDESS dataset using MFCC and spectral features (Librosa) on TensorFlow 2.10 with CUDA 11.2/cuDNN 8.1.
- Fused Wav2Vec2, DeepFace, OpenCV, and Transformers into a low-latency, GPU-accelerated pipeline.

SKILLS

Languages: Python, Java

AI / ML: PyTorch, Scikit-Learn, HuggingFace, LSTM, LLM Fine-tuning, NLP, OCR, VLMs

Computer Vision: OpenCV, MediaPipe, BlazePose, CVAT, Object Detection, Image Segmentation

Development, Tools & Tech: Flask, MySQL, MongoDB, Node.js, React.js, Git, GitHub, VSCode, CUDA, RunPod (Cloud GPUs), Unity 3D

Coursework: Data Science, DBMS, Data Analytics, Machine Learning, AI, Python, Java

ACHIEVEMENTS

- **Patent:** “A Virtual Body Augmented with AI Assistant” — App No: 202211074491
- **Hackathons:** Led teams in **7 Hackathons**; SIH 2024 Finalist; Vihaan 8.0 (DTU) Finalist.
- **Workshops/Training:** Student Placement Coordinator; AI/ML (IIT Bombay); Python (IIT BHU).
- **Certification:**
 - British Council English: 577/600 (C1 Advanced).
 - Pursuing IIT Madras BS degree in Data Science. Passed Foundation Level, currently in Diploma Level.